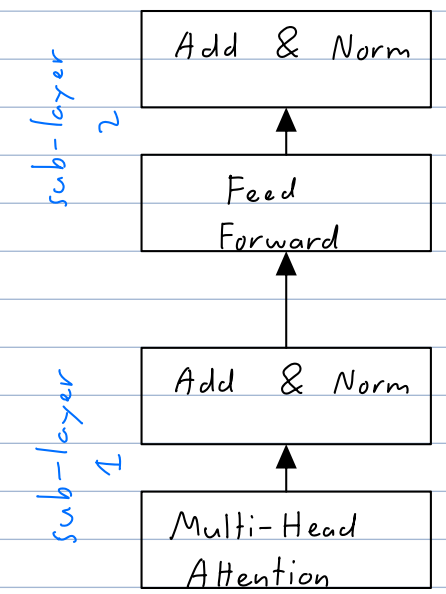
This is about how the pieces come together into a repeatable unit (block) and stacked.

A transformer block:

- Sub-layer 1 — Attention, where each word gathers information.
- Sub-layer 2 — Feed-forward, where each word process information.



The 2 helpers:

Helper 1 - Add (Residual connection)
- Adds the change produced by the step onto the original word.

Helper 2 - Norm (Layer Norm)
- After addition, adjust the 512's numbers to normal size.
- Stops them from spiralling too big or small, to keep the model stable.

Masking

Masking is a rule that blocks each word from looking at any word that comes after it. So the model is able to predict the next word.

The setup:

1. Each word makes a query and a key
2. Every query is scored against every key
3. Softmax turns those scores into weights that add to 1
4. Each word pulls in a blend of the others.

Masking occurs between step 2 and 3. It sets all future scores to $-\infty$ before softmax, to return a value of 0.
Softmax takes scores and turn them into weights, via exponentiating to the power of each score:

$$e^{-\infty} =$$